Supplementary Material

Open-Vocabulary 3D Scene Graphs from Sparse Views via Duplicate-Aware Fusion and a De-Circularized Benchmark

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This supplement holds material referenced from the main paper: the open-vocabulary detection front-end on real frames (Fig. S1), the duplicate-merge IoU-threshold robustness sweep (Fig. S2), the two detailed per-class tables (Tables S1 and S2), the full uncertainty-gate negative result (Table S3), and the per-view duplicate-aware gains with per-scene win counts (Table S4). All numbers match the released per-scene scores; the headline values are already stated in the main paper.

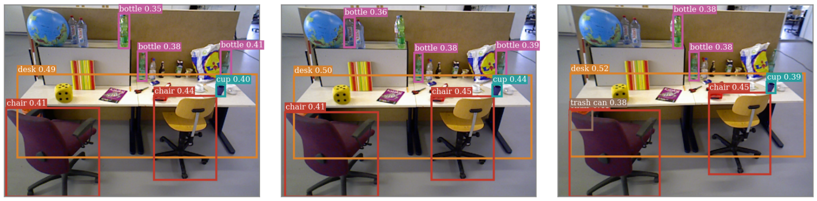

Fig. S1. The open-vocabulary detection front-end on real frames (three views of `freiburg3_long_office_household`). OWLv2, queried with the indoor vocabulary, returns named, boxed objects (desk, chair, bottle, cup, trash can) with confidences, the property the lifting and fusion stages need; the common mask-then-CLIP alternative returned room-scale category noise on these blurry frames rather than the objects present.

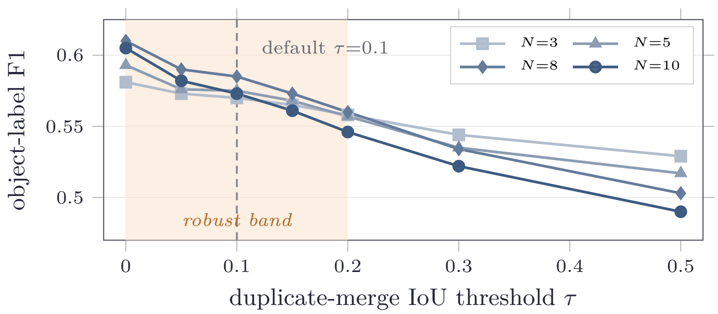

Fig. S2. Robustness of duplicate-aware fusion to the merge threshold τ (object-label F1, mean over the 28 scenes, at N=3/5/8/10 views). The gain over GRAPH-FUSION is flat across the low-IoU band $\tau \in [0, 0.2]$ (average Δ from +0.118 at τ=0 to +0.075 at τ=0.2) and fades only once τ is strict enough to stop merging duplicates (+0.054 at τ=0.3, +0.030 at τ=0.5). The default τ=0.1 (dashed) sits within noise of the optimum.

Table S1. Per-class characterization of GRAPH-FUSION (micro-averaged over 28 scenes, dropping `unknown object`). Recall rises and precision falls as views are added; F1 is best at the sparse end. The over-counting signature behind the main-paper finding.

Views	Precision	Recall	F1
3	0.519	0.700	**0.596**
5	0.429	0.739	0.543
8	0.361	0.794	0.496
10	0.331	0.802	0.468

Table S2. Per-class mechanism of duplicate-aware fusion at v8 (micro-averaged over 28 scenes). Dedup recovers precision on the over-counted classes while preserving recall for almost all classes; the only recall cost is on closely-packed same-class instances (`desk`, `bottle`).

Class	GRAPH-FUSION P/R	dedup P/R	Δ recall
`monitor`	0.43 / 0.95	0.64 / 0.95	0.00
`chair`	0.45 / 0.91	0.71 / 0.91	0.00
`floor`	0.27 / 0.93	0.71 / 0.93	0.00
`wall`	0.26 / 0.69	0.60 / 0.69	0.00
`desk`	0.15 / 0.96	0.45 / 0.92	−0.04
`cabinet`	0.36 / 0.27	0.67 / 0.27	0.00
`keyboard`	0.39 / 1.00	0.50 / 1.00	0.00
`person`	0.79 / 0.79	1.00 / 0.79	0.00
`bottle`	0.70 / 0.76	0.78 / 0.72	−0.04

Table S3. Uncertainty-aware fusion is a negative result. Δ PROPOSED − GRAPH-FUSION in object-label F1 (rank-normalized, w=0.3); W/L/T is per-scene over $n = 28$. A weight sweep over $[0.1, 0.8]$ contains no setting that beats the baseline.

	v3	v5	v8	v10
Δ PROPOSED − GRAPH-FUSION	−0.033	−0.053	−0.082	−0.096
Per-scene W/L/T	2/22/4	0/26/2	0/26/2	0/27/1

Table S4. Duplicate-aware fusion versus the GRAPH-FUSION baseline and the FIXED-SHRINK control. Δ is the object-label F1 gain; W/L/T is per-scene wins/losses/ties for dedup over GRAPH-FUSION ($n = 28$, sign test $p < 0.001$ at every view).

	v3	v5	v8	v10
Δ dedup − GRAPH-FUSION	+0.057	+0.083	+0.120	+0.122
Δ dedup − FIXED-SHRINK	+0.065	+0.101	+0.141	+0.154
Per-scene W/L/T (vs GRAPH-FUSION)	22/1/5	25/0/3	27/0/1	26/1/1